

Review article

Wired, wild, wonderful: A scoping review of early childhood nature connections fostered by digital technologies

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ABSTRACT

Nature connections in early childhood are being lost due to increasingly urban and managed lives. Yet, emerging research is signalling the usefulness of digital technologies in reconnecting children with the natural world. This review scopes the last ten years of research reporting child-nature-technology interactions to identify: (1) methods, (2) approaches to children's agency in research and design, (3) technologies used and the experiences they produced. After screening, a search of seven databases produced 24 papers that were analysed across these three dimensions. We describe the current research and make recommendations including more granular reporting of participation, increased engagement with theory and nature connection frameworks, and consideration of methods that encourage children's agency at earlier stages in design work. We also suggest novel directions when designing for nature experiences that encourage personal meaning-making, relationality, and build child competencies for a rapidly changing world.

1. Introduction

The urbanisation of modern life is creeping into childhood, with increasingly digitised entertainment and reduced interactions with nature. Louv's Last Child in the Woods (2005), was a call to action for a generation, sparking a renewed interest in children's nature play and the role technology plays in reinforcing sedentary, indoor lives. Human-computer interaction research also took up this call, with studies exploring a range of technology-nature interactions to better understand the ways in which these might be designed for. Yet of these, few dealt with early childhood, nor explored what it meant to design for and with children. In response, this review examines the role digital technologies can play in early childhood connection with nature, an emerging area of child-computer interaction research (Antle & Hourcade, 2022), to learn about methodological gaps and design opportunities. Importantly, this review also seeks to understand child agencies, how they are incorporated in research and design, and how these might be further encouraged.

Early childhood (birth to eight years) is a critical life stage for development and learning. Historically, children were often excluded from research and decision-making processes (Dobson, 2020; Greig, Taylor, & MacKay, 2007; Robinson & Diaz, 2016). This failure to recognize them as independent thinkers who have valuable insights to offer and a right to participate in decisions that affect their lives (UNCRC, 1989) resulted in a poor understanding of children's experiences and needs. Recently, a review of methods in child-computer-interaction showed that research on users' needs and prototype-building still rarely engages with early childhood participants (Tsvyatkov & Storni, 2019). Understanding early childhood-technology interactions is essential; firstly, because young children deserve a presence and ideally a voice in research; secondly, because understanding the early years can generate valuable insights for researchers, educators, families, and other stakeholders.

Young children's nature interactions are also influential on later life outcomes. The rich sensory learning opportunities available in natural settings are associated with pro-environment attitudes, as a sense of

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being connected to nature is tied to care for nature as an adult (Chawla, 2020; Shanahan & Fuller, 2016). Children's access to nature, including metropolitan green spaces, is also associated with a range of health benefits (Chawla, 2015). Nature-based learning is also reported to outperform traditional indoor instruction (Kuo, Barnes, & Jordan, 2019). Consequently, a range of child-nature engagement frameworks have surfaced to qualify child-nature connectedness in directing and assessing interventions (Chawla, 2020; Giusti, Svane, Raymond, & Beery, 2018; Harvey, Hallam, Richardson, & Wells, 2020; Lumber, Richardson, & Sheffield, 2017). However, few attempt to define 'nature'.

While what 'nature' is may appear self-evident, children increasingly experience urban lives and access different forms of nature than those in rural settings. Hence, we investigated the physical and theoretical contexts in which child-nature-technology research is conducted. Similarly, our focus on early childhood necessitated understanding the role adults play as facilitators. This is particularly important as recent research is suggesting that parents' orientation to nature is more impactful on young children's nature experiences than access to nature or screen time (Van Truong, Nakabayashi, & Hosaka, 2023). There was also a need to understand reported research regarding young children's engagement with nature and digital technologies, and to explore relations highlighted across different studies which could contribute to shaping children's experiences.

Consequently, our scoping review aimed to answer the following questions:

RQ1. What research is being carried out regarding child-nature-technology interactions, in terms of the settings, participants, methods and guiding theories and constructs?

RQ2. How are children's agencies facilitated by the research and/or design processes, and what role do adults play in supporting this?

RQ3. What kinds of interactive digital technologies are being developed or used to foster young children's (birth to 8 years) nature experiences and/or connection, and what kinds of experiences do these technologies provide?

Due to the recent emergence of designing for child-nature interactions, we include studies where design was not the aim, but instead technologies were introduced to children to aid the researcher in learning about their experiences and points-of-view. The benefit of this approach includes presenting research with devices that children are likely to have access to (e.g., digital cameras, smart phones, computers) and that can provide an understanding of basic interactions such as sensing and recording, as well as expanding the field of literature included.

We developed an overview of research from the last thirteen years, identifying gaps that could be used to shape future research. Specifically, we make recommendations regarding reporting, participation, and measurement of impact, as well as approaches to child agencies in research, and technology design.

2. Conceptualisations of nature and care

Understanding the research on child-nature connection demands understanding of the concept of nature itself. Soga and Gaston (2016), who introduced the term the 'extinction of experience', highlighted both the decline and importance of nature experiences. They make a distinction between nature as any experience with living organisms versus experiences in and with 'wild nature', with nature experiences having emotional, physical, spiritual, and intellectual dimensions (Gaston & Soga, 2020). Despite growing human-nature connection research, its conception is still fundamentally split and diverse (Ives et al., 2017). Care for nature - and care itself - therefore needs careful consideration, particularly when framed as a design goal.

'Care' frequents literature regarding nature connectedness and

environmental matters. To care can be considered a political project (Black & Dwyer, 2021), which calls upon people to engage in ethical situations in responsive and effective ways (Taggart, 2016). Critically, matters of care extend beyond humanity and towards caring for the environment. For instance, Puig de la Bellacasa writes: "Care is a human trouble, but this does not make of care a human-only matter" (2017, p. 3). The intersections of care and nature connectedness have been explored via research framed through a common worlds ethic and focusing on the more-than-human world (e.g. Hodgins, Kummen, & Merewether, 2022; Land et al., 2020; Merewether, 2020), with emphasis on exploring how we share the world with ecosystems, flora, and fauna (e.g. Häggström, 2019; Renshaw, Jackson, Mortlock, & Tooth, 2023).

3. Children's nature connectedness

The reasons for child-nature connection being fundamentally threatened are diverse, including increased urbanisation (Soga, Yamanoi, Tsuchiya, Koyanagi, & Kanai, 2018; Turner, Nakamura, & Dinetti, 2004), which potentially drives parental control due to fears for children's safety (Clements, 2004), and also the increasing accessibility of screen-based entertainment in the home. However, recent research suggests that family members' attitudes to nature may be more impactful than screen-time (Van Truong et al., 2023), and this is likely to be more so for very young children who depend upon adults for play and learning experiences. Hence, understanding children's nature connections also requires understanding of the adults' facilitation role.

Various frameworks attempt to describe and assess child-nature connections from multiple perspectives. For example, the 'Assessment framework for Children's Human Nature Situations (ACHUNAS)' presents phases that reflect how children progress in their connectedness: "being *in* nature, being *with* nature, and being *for* nature" (Giusti et al., 2018). Another, Harvey et al.'s (2020) extended framework, provides a flexible structure for practitioners to follow when developing activities to engage children with nature. The framework combines 'pathways to nature connectedness' (i.e., emotion, contact, beauty, meaning) (Lumber et al., 2017) that are associated with the biophilic values proposed by Kellert (1993); the "good things" children notice in nature. Harvey et al. (2020) explored the "good things" through a daily positive psychology writing intervention. Based on this intervention, the developed framework highlights ten themes of positive things children notice in nature, and five micro-activities per theme. Other practical activities are suggested in Chawla's (2020) review of studies of children's nature connection and coping with environmental loss. Ways to build this connection include access to nature, with meaningful respect to age, gender and family relations, as well as positive experiences. These experiences include making recordings through science inquiry, creative writing, and the arts, and joining collective initiatives which contribute to having a sense of confidence and agency in nature. Chawla (2020) reports that nature connection experiences were fostered by learning about actions to take, and supported by seeing peers and kindred others care for nature.

Nature connectedness is considered a determinant of sustainable behaviours towards nature (Barrera-Hernández, Sotelo-Castillo, Echeverría-Castro, & Tapia-Fonlem, 2020). Childhood is recognized as a time to build emotional connections to nature, which can potentially develop into care and respect (Ives et al., 2018). This suggests that children who experience greater connectedness will embrace more sustainable behaviours. Additionally, Chawla (2020) notes that while most child-nature connection studies focus on feelings of comfort, competence, and enjoyment in nature, the current global environmental threats can also generate feelings of sadness or anxiety. The review ends with a suggested integration of practices that encourage positive experiences and coping with environmental change, to help children connect with nature and develop constructive hope and action.

On the point of assessing nature connection in early childhood, few validated self-report measures exist, and methods include surveys in the

higher ages (Larson, Green, & Castleberry, 2009), observation, and reporting by parents, as well as various interview measures and techniques (Chawla, 2020), including biophilia interviews using puppets (Rice & Torquati, 2023). Interviewing children appears to be the main assessment method in early childhood (Beery, Chawla, & Levin, 2020). With recent digital interventions in children's nature connection, how well their effectiveness is assessed remains an open question.

4. Designing technology for nature connectedness with and for children

Digital technology seesaws in the public imagination in terms of its impacts on children's lives. Its mixed potentialities include the negative impacts of engagement with the attention economy and indoor, sedentary lifestyles, along with positive impacts, such as engagement in citizen science for environmental conservation (Preece, 2017), and building STEM skills, such as through creative coding (Murcia, Pepper, Joubert, Cross, & Wilson, 2020). In parallel, a movement within design research focused on environmental and social care is producing technologies and design methods seeking to intervene in nature connection. These range from Heads Up Games, which encourage children's outdoor play without screens (Soule, Markopoulos, & Magielse, 2010); ambient technologies that tie to ecological science initiatives (Dema et al., 2020); design through exploring the observation of cyclic natural phenomena (phenology) and human-nature relations (Rodgers, Ploderer, Vella, & Brereton, 2023); and augmented nature for outdoor learning (Rogers et al., 2004), amongst many others. Studies of both commercially available and DIY nature cameras also suggests directions for using passive sensing to increase urban families' awareness of nature in their backyards (Gaver et al., 2019; Vella, Ploderer, & Brereton, 2021). This last point suggests directions around appropriating accessible technologies for building nature connection.

Ways of framing technology design for nature connection are suggested by Webber, Kelly, Wadley, and Smith (2023). The authors speak of six roles that technology can play in nature connection, including: revealing and familiarising people with the natural world; pooling knowledge (e.g., citizen science); producing 'nature on demand' for human wellbeing; learning about nature; discovering and navigating nature sites; and experiencing nature from other perspectives. An earlier review focused on children and environmental HCI found three main types of technology use - instructional, exploratory, contributory - and called for the design of technologies that encourage care for nature on its own terms, and finding value beyond that of human sustainability (Anggarendra & Brereton, 2016). As we consider the evolving ways of approaching technology design, the voice of the user as co-creator becomes a pertinent paradigm shift to contemplate.

Co-designing technology with children enables their active input, harnessing their unique perspectives, experiences, and values while affording agency and voice. Children's roles in technology design vary from user, tester, informant, to design partner, in which they have opportunity to contribute throughout the design process (Druin, 2001). Participatory methods used to learn about children's perspectives on nature and the outdoors include photovoice (Adams, Savahl, & Fattore, 2017), wearable cameras (Cumbo & Leong, 2015), fictional inquiry and co-envisioning using 'magical layers' (Cumbo et al., 2014, 2020), design probes and storyboards (Om, Brereton, Dema, & Ploderer, 2021), speculative design in the form of magic machine workshops (Andersen, 2013), and game co-design (Vella, Dema, Soro, & Brereton, 2021), amongst others. Some of these explorations have determined the importance of designing for children's imaginative interactions with nature (Cumbo et al., 2020; Om et al., 2021).

Imagination also forms a bridge to understanding how to design for children's creativity. The role of creativity in children's digital play has been explored by Murcia et al. (2020). Through a study focusing on three- and four-year old's engagement with digital coding technologies, the 'A to E of Creativity' framework was developed, with clusters

comprising agency, being curious, connecting, daring and experimenting (Murcia et al., 2020). While technologies that facilitate children's nature connection through creative approaches are few, directions are indicated by interfaces that mesh with children's existing creative practices, such as the FingerDraw digital paintbrush (Hettiarachchi, Nanayakkara, Yeo, Shilkrot, & Maes, 2013).

Similarly, understanding what directs children's attention is vitally important for child-nature-technology relations. An important consideration is that children not have their attention consumed by technology interactions when the aim is nature connection (Om et al., 2021). Research coming from the design space of outdoor games, suggest three major approaches including pervasive games (e.g., location-based interactions); Heads Up Games using tangibles, and importantly no screens in outdoor settings; and transparent technology, in which digital devices are indistinguishable from objects already used outdoors (Hitron et al., 2018). However, given that these games are typically designed to inspire physical activity it is unclear how nature connection is influenced. Alternative ways of attracting attention to nature may include augmenting natural features, such as with the ListenTree, which produces sounds through bone conduction when the listener leans their head against the tree (Portocarrero, Dublon, Paradiso, & Bove, 2015). The following seeks to understand these nature-technology interactions through the lens of early childhood.

5. Method

This study aims to outline gaps in approaches to designing for children's nature connection. To achieve this we conducted a scoping review, informed by methods outlined in Pollock et al. (2023). The research consisted of these stages: search, screening, data extraction, and analysis, which are described in the following sections.

5.1. Search

The search strategy aimed to locate peer-reviewed published studies. An initial search on ACM Digital Library using the Extended Library Collection was conducted to identify keywords with this later expanded to other databases. The words contained in the titles and abstracts of relevant articles were used to develop a full search strategy for the following databases:

- Web of Science
- Scopus
- ProQuest Eric and Education databases
- ACM
- IEEE
- Taylor & Francis
- Wiley

English-language studies were included. Studies published since 1 January 2010 were included as this is the year that the iPad was released (a popular interactive digital technology used across contexts with young children); consequently, over 13 years of research are encompassed. We included search terms related to children aged birth to 8 years and the educational contexts they may frequent; terms related to nature experiences and connection; and interactive digital technology terms. We excluded terms that may have netted studies with students outside of our age range. The complete search string is provided in Appendix A. This was adapted minimally to comply with different database requirements. The first search was concluded on 3 March 2023. A second search encompassing literature from that date until end May was also conducted.

5.2. Screening

Following the search, all identified citations were collated and

uploaded into Endnote 20 and duplicates removed. Potentially relevant studies were retrieved in full and their citation details imported into the JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI) (Piper, 2019). After a pilot test, titles and abstracts were screened by two independent reviewers for assessment against inclusion criteria. Decision trees for both the abstract and title and full text screening stages are provided below (Fig. 1). Importantly, we included papers in which the child participants used any kind of digital technology, whether this was designed for nature connection or not. This allowed us to learn more about how commercially available technologies may be integrated into digital systems and the prevalence of different technological affordances.

All screening (abstract and title, and full text) was carried out by two independent reviewers from the first five authors listed against inclusion criteria drawn. Initial title and abstract screening across 1541 citations produced 157 papers for inclusion in the full text screening.

A second search was conducted that used the same databases and terms to collect records published between the dates of 4 March - 31 May. An additional 33 papers were abstract screened, with only 6 papers continuing to full screening, resulting in 163 records included for screening. Reasons for exclusion of records at full text that did not meet the inclusion criteria were recorded. Any disagreements that arose between reviewers at each stage of the selection process were resolved through discussion, with additional reviewers' input.

126 papers were excluded at the full text stage for the following reasons (number of papers are reported in round brackets):

- Body of the paper not in English (4)
- Published before 2010 (1)
- Duplicates (3)
- Full text could not be retrieved by either of our institutions (4)
- Paper was not peer-reviewed (5)
- Study was a review of the literature and not a primary source (10)
- Technology reported in the study was not a digital technology (2)
- Study did not have children as participants (11), or did not include children aged birth to 8 years (32)
- No focus on children's nature engagement or experiences (11)

- No information included about children, nature engagement, and technology (13), or didn't measure or study the effect of the child participants' technology use on their nature engagement (16)
- Technology did not have a relationship to the children's nature engagement (14)

This resulted in 37 papers for data extraction. We examined the bibliographies of the included papers and added two papers to the corpus, bringing it up to 39 papers.

Deeper reading during data extraction resulted in a further 15 papers being excluded due to: unclear child participation (3); lack of digital technology (3); no relationship between technology and children's nature engagement (1); no nature engagement (2); unclear ages of children (1); and duplicate studies (5).

Studies that were duplicated were compared against the amount of extractable information and included if they provided more information. Three papers refer to the same three technologies, but rather than include just one paper, all three were included to capture the most information about the design and research methods used for each digital technology, respectively Eko (Caiola et al., 2023), ABBOT (Delprino et al., 2018), and GAIA (Gennari et al., 2019).

5.3. Data extraction

A data extraction tool was created outside of JBI Sumari using Google Forms, which allowed for greater field customisation. The data extraction tool was iteratively developed amongst the five reviewers and supplied multiple choice and free text items. The tool is provided in Appendix B. In brief, it collected information on aims, methodology, methods, theories and constructs engaged with, participant details, research context (country), types of nature, details about the technology the children used and the resulting types of nature experiences, how much agency the children had in terms of technology use and their role in the research, and the role of adult participants.

Data extraction was carried out by one reviewer per paper. Issues were discussed between the first five authors, resulting in further exclusions, as noted above. The final papers included for analysis totalled

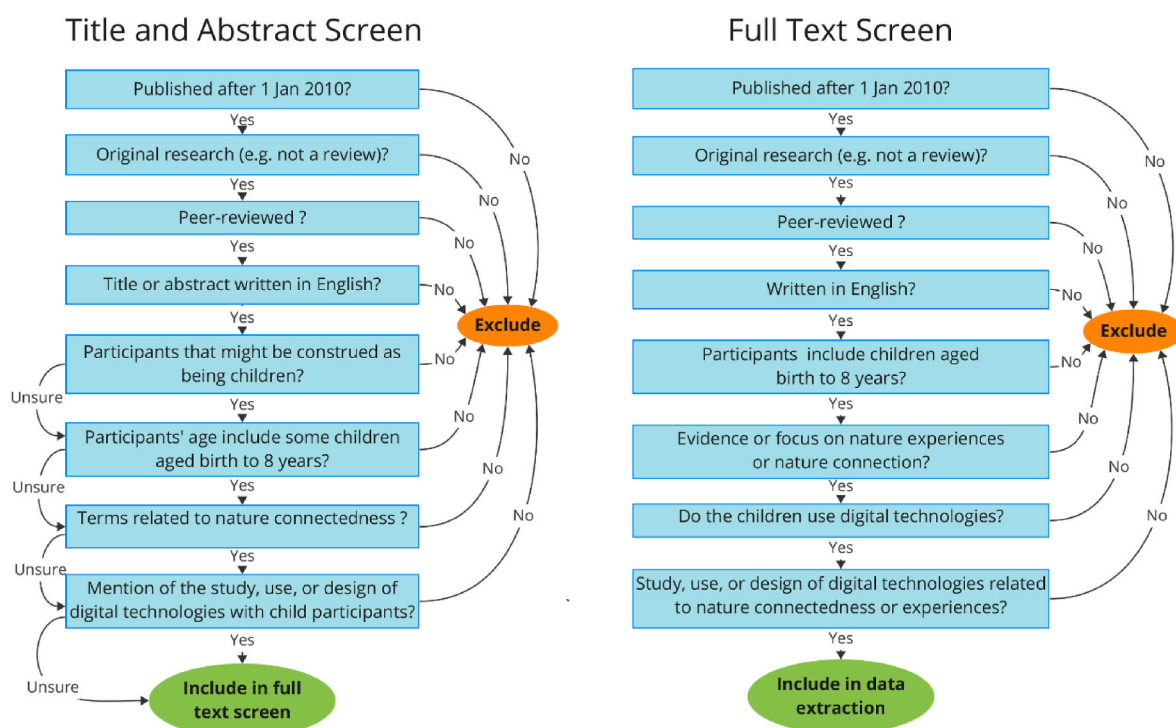


Fig. 1. Screening decision tools.

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5.4. Analysis

Analysis was undertaken by the first four authors, with regular discussion to ensure continuity of analysis and agreement between the team. Fields that contained categorical data were synthesised in Excel (Microsoft Corporation, 2018) and provided as figures. Fields from the data extraction sheet that contained qualitative data were imported into Miro (Chan, Ho, & Tom, 2023) as sticky notes. These were subject to analysis through descriptive qualitative techniques (Peters et al., 2020).

The analysis was framed by the authors' interests in participatory and child-centric research and designing for human-nature relations. The authors range from early career to established academics in the fields of human-computer interaction and education.

6. Findings

The findings are broken down into three major sections, addressing each research question in turn. Section 6.1 describes the research being carried out (RQ1), Section 6.2 characterises how child agencies and the role of adults are enacted in the research (RQ2), and Section 6.3 describes the kinds of digital technologies children are using and their subsequent nature experiences (RQ3).

When referring to statistics drawn from the body of literature in scoping review, the number of papers are reported in round brackets, e. g. (5). When numbers are reported in this way, see Table 1, Appendix C, for detail linking these findings to references. All reference to 'technology' refers to digital technologies.

6.1. Types of research engaging with child-nature-technology interactions

6.1.1. Where and when the research was carried out

The year of publication (Fig. 2) shows growing interest in the field until 2018. There was a drop in 2019–2020 which might have been caused by pandemic impacts on research. However, the maximum number of studies is found in the year 2021 (6). Thus, there may have been a delay in 2019 and 2020 that can be resumed in 2021. We anticipate more publications in 2022–2023 becoming available after our search period. Based on the trend apparent, it is likely that publications in the field will continue to increase.

The most common research location was the United States. Only one study was carried out across two countries: Australia and Canada (Land et al., 2020). See Fig. 3, and Table 1, Appendix C, for the linked studies.

Most studies focused on human-managed spaces with natural features (18) including contexts such as gardens, parks, playgrounds, botanical gardens, arboreta, school grounds, and university campuses. A smaller proportion were focused on wild nature outside of urban settings (9) including forests, intertidal sites, and moorlands.

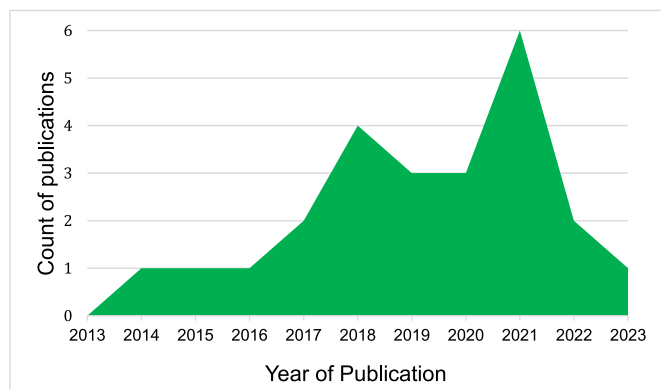


Fig. 2. Dates of publication.

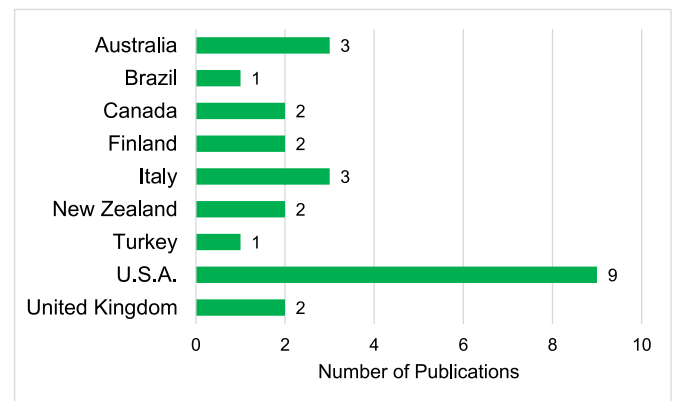


Fig. 3. Countries the research was carried out in.

Fewer studies focused on wild nature in urban settings (5) and only one reported research being carried out on farmlands (White, 2015). One paper did not clearly specify context though the technology was used by educators on field trips and the data collected was of natural features (Rikala & Kankaanranta, 2014). Nine papers carried out research across multiple settings. See Appendix C for detail.

6.1.2. Research aims

Study aims were broken down across two main methodological approaches: to carry out an evaluation of a technology designed for nature connection or experiences, or to highlight and learn from the voices of children (see Appendix C, Table 1). Within the former, evaluations were largely situated around design research (see breakdown in Section 6.2), with two of these then used to develop frameworks for technology design (Caiola et al., 2023; Kawas, Chase, Yip, Lawler, & Davis, 2019). Outside of this, two papers respectively evaluated a larger program for enhancing environmental citizenship through citizen science, in which school-based technology played one part (Carson, Rock, & Smith, 2021); and the use of nature photography as a means for supporting academic skills development (Chambers, 2018).

Papers that prioritised children's voices were found to focus on their nature experiences and what they valued, to inform other topic areas. These topics included place and placemaking (Bertagnoni & Galheigo, 2021), and subsequent interrelations with technology (Land et al., 2020) and pedagogy (Sesigur & Edeer, 2021). Pedagogical impacts of nature experiences were typically explored using handheld and GoPro cameras and focussed on understanding how to engage with outdoor learning generally (Bates, 2020) and for environmental education specifically (Harwood, Barratt, & Collier, 2019).

Learning about children's nature experiences also informed:

- Learning about their environmental identity development (Green, 2017);
- Child-family-nature relations (Green, 2022; White, 2015);
- Subsequent climate-change related political action (Jones, 2022).

6.1.3. Methods

Methods were primarily qualitative (18), with an additional five studies using mixed methods (Caiola et al., 2023; Carson et al., 2021; Gennari et al., 2019; Kawas et al., 2019; Paulsen, Carroll, Paulsen, & Andrews, 2021), and one study presenting a purely quantitative analysis (Bruni, Winter, Schultz, Omoto, & Tabanico, 2017). In terms of design methods, a range of approaches to data gathering were apparent. A majority of studies included interviews with participants (17); at times, participant experience and perspective was also explored via observations (13), workshops (Bates, 2020), and conversations that formed part of a collaborative approach to meaning making (Land et al., 2020). Many studies included participatory photography and videography (9),

with some using photovoice (Bertagnoni & Galheigo, 2021; White, 2015). These collections formed part of the dataset, along with written artefacts such as text that accompanied photographs (Jones, 2022) diaries used by educators (White, 2015), family members (Soro et al., 2018), and researchers (Sesigur & Edeer, 2021), and digital stories (Kumpulainen, Byman, Renlund, & Wong, 2020; Renshaw, Jackson, Mortlock, & Tooth, 2021). At times, researchers utilised their own reflections (Caiola et al., 2023; Sesigur & Edeer, 2021). Some studies used questionnaires (Bruni et al., 2017; Caiola et al., 2023; Carson et al., 2021; Gennari et al., 2019; Jones, 2022), with one administering a game version of the Impact Association Test (Nature) (Bruni et al., 2017). Six studies engaged child-led tours of natural environments (Bates, 2020; Bertagnoni & Galheigo, 2021; Green, 2017, 2022; Harwood et al., 2019; White, 2015) using video and/or audio documentation.

In terms of participants, reporting of age varied across papers. At times school grades were referenced without further explanation as to the age-bracketing relevant to the research context; for example, it was stated that samples included or comprised “preschool and kindergarten-aged children” (Land et al., 2020), “first graders” (Zydney & Schaen, 2018), and “groups of students in grades 3–5” (Chambers, 2018) without further specific elaboration. In keeping with Fig. 4, when ages were stated, these predominantly fell in the range of seven to eight years of age (18). Fewer studies included children aged five to six years (11) and three to four years (7), and none included children from the birth to two years bracket. Several studies worked across age brackets (9), with most of these in the 5–8 years bracket (6). See Fig. 4, with Appendix C, Table 1 providing a breakdown across papers.

The ethnicity of children was only reported in seven studies (Chambers, 2018; Green, 2017; Kawas et al., 2019, 2021; Kumpulainen et al., 2020; Land et al., 2020; Paulsen et al., 2021) with some diversity apparent. One study provided the following acknowledgment: “We want to make public that the educators and researchers, and the great majority of children, who participate in this project are settlers to these places. While we have diverse identities and relations within settler colonial flows, each of the educators and researchers come to the Facetimeing project with an ethic of addressing settler accountabilities” (Land et al., 2020). One study focused on Indigenous participants (Green, 2017), while several others reported diverse representation (Chambers, 2018; Kawas et al., 2019, 2021; Kumpulainen et al., 2020; Paulsen et al., 2021). The extent of this representation varied - for instance, one study reported that their sample swung to middle-/upper-class participants, but that it mirrored the race distribution apparent in the city where the study took place (Kawas et al., 2021). Two other studies emphasised diverse representation apparent in their participant sample; for example, in one instance it was apparent that ethnically and socioeconomically diverse children were recruited (Kawas et al., 2019) and in another, it was evident that the sample included children with special educational needs and with diverse linguistic backgrounds (Kumpulainen et al., 2020).

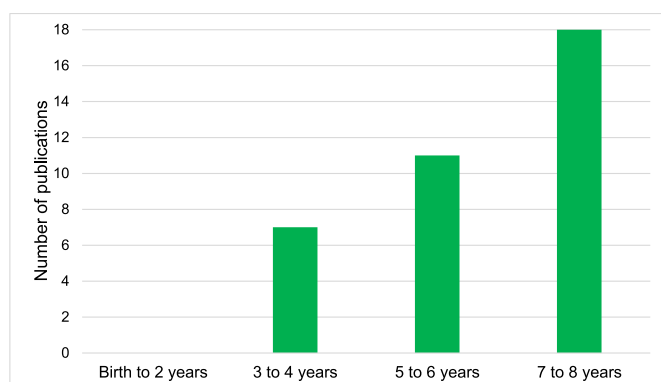


Fig. 4. Distribution of age brackets across the papers.

6.1.4. Theories and constructs

Several papers engaged with theories and constructs to guide the research with children, and oftentimes the analysis. Papers that engaged with relational theories and constructs were grounded in new materialism, and/or posthumanism (Kumpulainen et al., 2020), as well as a common worlds ethic and framework (Harwood et al., 2019; Land et al., 2020). Papers also engaged with constructs to understand or measure levels of nature connectedness including the Implicit Association Test (Nature) (Bruni et al., 2017), and Louv's 'nature deficit-disorder' construct (Caiola et al., 2023). One paper described a citizen science project engaged with environmental citizenship (Carson et al., 2021). The remaining papers presented a diverse set of constructs with no immediate connection to nature connectedness (e.g. visual surplus, White, 2015) drawn from fields such as psychology, sociology, and education.

'Nature' was only defined in three papers. Two of these papers engaged strongly with relational ontologies, with one defining nature as “a system of material and performative relations that include physical and material environments” (Kumpulainen et al., 2020, p. 3), and the other acknowledging “interrelations among/between the human and more-than-human world” (Harwood et al., 2019, p. 58). The third paper engaged less with theory and more with aesthetics, and framed nature in terms of its how it has been appropriated in pedagogical contexts to be “detached from its living context (a physical and spiritual experience)” (Sesigur & Edeer, 2021, p. 98).

6.2. Child agencies in research and design

Children's agency in the research and design processes varied, as did its reporting, but overall, there was a trend towards child-centred methodologies. Eight papers reported on the participatory design of digital technologies, whether engaging children as testers or users or involving them throughout the design process (Caiola et al., 2023; Delprino et al., 2018; Gennari et al., 2019; Kawas et al., 2019, 2021; Kumpulainen et al., 2020; McClain & Zimmerman, 2016; Soro et al., 2018). One paper reported the use of digital technology to aid the codesign of an outdoor learning space (Bates, 2020). Children's early engagement in the design of digital technologies was only documented in a few cases (Caiola et al., 2023; Delprino et al., 2018; Gennari et al., 2019; Kawas et al., 2019). Within papers that aimed to learn more about children's nature experiences, only five demonstrated children leading or providing direction about activities (Bertagnoni & Galheigo, 2021; Green, 2022; Jones, 2022; Sesigur & Edeer, 2021; Zydney & Schaen, 2018).

At times, children were afforded creative agency, with eight papers speaking to methods where children were involved in creative production and expression. Namely, children were engaged in photography where they had agency over what photos were taken, selected, and/or shared (Bertagnoni & Galheigo, 2021; Chambers, 2018; Jones, 2021, 2022; Land et al., 2020; Renshaw et al., 2021; White, 2015), storytelling (Kumpulainen et al., 2020), and the creative editing of their data gathering (Land et al., 2020). In many of these studies, choice and agency was emphasised; for example, they were encouraged to engage in photography “without any further provocation... as adults did not seek to influence or restrict the children's aesthetic choices within their photography” (White, 2015) and with a focus on discovery of nature (Chambers, 2018) and “enjoying looking and taking pictures of whatever caught their attention” (Jones, 2022), or in some cases, what was most meaningful to them (Bertagnoni & Galheigo, 2021).

Some studies integrated approaches that enhanced children's agency, such as member-checking (Jones, 2022) and emphasising participatory agency. For instance, in some studies where wearable cameras were utilised, children were encouraged to move freely, follow their interests, and determine the span of their use of the wearable camera. For example: “Children choose if, and when they would like to wear a camera and for how long. When a child is finished wearing the camera they simply tell the researcher” (Green, 2022, p. 20).

In other studies, child agency was variable. Where adults were fellow participants (e.g., parents, family members), agency ranged depending on whether adults permitted children to lead or whether they exerted greater control (Paulsen et al., 2021). There were also studies where agency was more limited, especially where technology and the use of it was prescriptive. For example, children were sometimes limited to how many photographs they could take: “They are very good at only taking one photo each, as per my instructions!” (Bates, 2020, p. 372). Technology also impinged on agency: “... according to the teachers there were various technical problems during the experiment which were difficult to solve alone [which] reduced the teachers’ desire to use application [and] also led to the situation where the children were not able to use the application independently” (Rikala & Kankaanranta, 2014, p. 176).

Notably, one study focused on an interdependent view of agency between the children and technology: “... neither the GoPro nor the child (or stick, thistle, snow, plastics, and so on) have agency on their own. Rather, it is the interplay between GoPro (matter) and the child that we argue is significant... the GoPro tips and pivots downward, the child readjusts her body in relation to the snugly bound camera about her chest, she uses a gloved hand to wrestle the GoPro back to its original position, the red light steadily blinks to cue the other children watching that the GoPro is functioning again” (Harwood et al., 2019, p. 60).

6.2.1. Role of adults

Adults played four key roles in the papers reviewed including leading children outdoors, facilitating technology use, educating children related to pedagogical experiences, and providing data to researchers. In most of the papers (17), adult participants, especially parents, helped in taking children to natural spaces. In ten papers, adults guided and facilitated technology use, such as trialling technology with children (Soro et al., 2018) and helping children use applications and websites (Bruni et al., 2017; Rikala & Kankaanranta, 2014). In one case, parents’ support of children’s data-logging led to adults’ involvement in the citizen science project (Carson et al., 2021). Seven papers reported participating adults educating children or linking experiences to pedagogy and five papers showed adults providing data to researchers about children’s experiences. While few papers reported more than one role for the adults, one study included adults in all four aforementioned roles (Rikala & Kankaanranta, 2014). One paper did not reflect on the role of adults except to gain consent for child participation (McClain & Zimmerman, 2016). Only one study focused on how adults and children negotiated digital technology use, and notably, this one also collected data on home use (Kawas et al., 2021). See Table 1, Appendix C for this detail linked to publications.

6.3. Digital technologies and the nature experiences they facilitated

The technologies discussed in this review served two purposes: first, to enable researchers to learn about children’s experiences, their interests, and the role of technology in those interactions; and second, as intentional interventions to enhance children’s connections with nature. The former was dominated by camera use whether handheld or worn. Only one paper in this category (Green, 2022), interrogates another form of technology in relation to children’s nature connection, and it was an electronic game embedded in a park, a Heads Up Game (Southe et al., 2010) that encouraged children’s physical activity. The remainder were either prototype tangible technologies; software using accessible devices (e.g., iPads, smartphones, and computers; or systems that combined these. See Fig. 5.

6.3.1. Technological affordances

The types of technologies used by children in this review primarily engaged users with nature through two senses: sight, and to a much lesser degree, sound. Only six papers reported on technologies that did not engage children in photography or video recording (Caiola et al.,

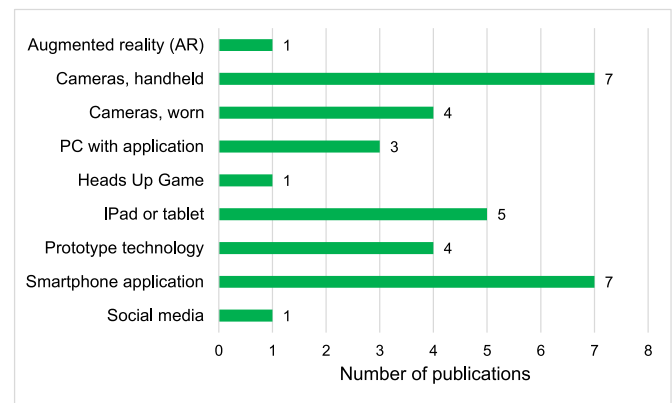


Fig. 5. Types of digital technology.

2023; Carson et al., 2021; Gennari et al., 2019; McClain & Zimmerman, 2016; Paulsen et al., 2021; Soro et al., 2018). Sound was engaged with via recording nature sounds (e.g., Caiola et al., 2023), and listening to nature sounds attached to videos (e.g., Soro et al., 2018). When digital technologies that engaged both children’s vision and listening were combined with storying or narratives, it provoked rich, meaningful, multisensory nature experiences (e.g., Kumpulainen et al., 2020; Renshaw et al., 2021). Combined sight, sound, and tactile engagement through technologies was apparent, consisting of several tangible devices through playful and gameful experiences (e.g. Delprino et al., 2018; Gennari et al., 2019; Soro et al., 2018), including the Heads Up Game that Green (2022) comment on.

Cameras were introduced to children for data collection to benefit researchers’ knowledge, but this was also an affordance of many technologies designed as an intervention. Of these, photography dominated as is noted above, however, sound collection was more recently explored (Caiola et al., 2023), as was text data-logging, with support from adults (Carson et al., 2021). Being able to manipulate this data to make new meanings or learn (e.g., filter, link, or creatively change) was an affordance offered by far fewer technologies. Only one technology focussed entirely on data manipulation, with this being in an indoor context (Carson et al., 2021).

Unsurprisingly, most technologies were intended (whether by design or researcher prompting) to be used outdoors. However, several were intended to bridge movements between outdoor and indoor activity, for example by encouraging data collection outdoors and data manipulation indoors. Only three were explicitly designed for indoor-only use: the Ambient Birdhouse, which intended to build knowledge of and sensitise users towards bird calls (Soro et al., 2018); a virtual hike of nature trails (Bruni et al., 2017); and archiving and analysis software associated with a citizen science project (Carson et al., 2021).

6.3.2. Nature experiences

The studies reviewed demonstrate rich experiences resulting from child-nature-technology interactions; see mapping in Fig. 6. Directional relationships cannot be presumed from the given information and so are presented as bi-directional. Despite this map suggesting that some experiences interact and not others, it seems likely that more relationships may result as this field of research expands. Within these 24 papers, the themes presenting most strongly include social experiences, aesthetic experiences, noticing, and care. The experiences provided can also be seen as due to project framing, e.g., citizen science objectives, and analytical approaches, e.g., relational analyses.

Children engaged and collaborated with family, friends, and classmates through various nature experiences enabled by the technologies. For instance, imaginative experiences were enjoyed among families where children often led or took turns in leading the experience, then shared these experiences through technologically mediated social

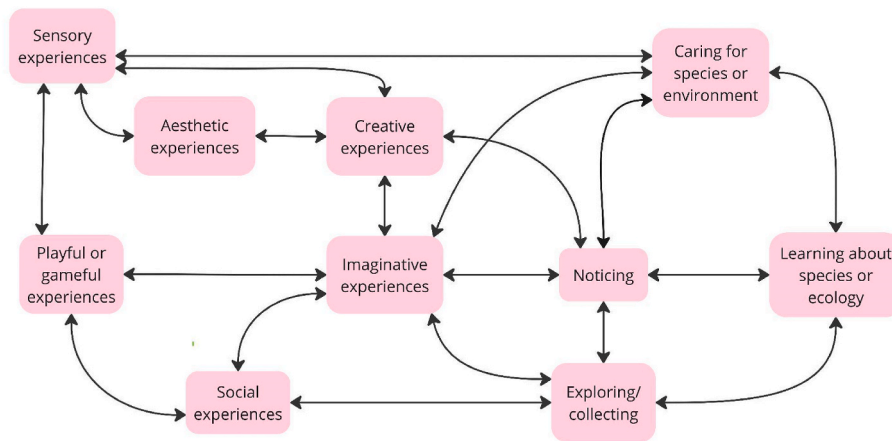


Fig. 6. Experiences produced by child-nature-technology interaction.

channels with classmates, teachers, and parents of other children (Renshaw et al., 2021). Children worked together in constructing virtual stories while exploring nature together (Kumpulainen et al., 2020). Some technologies were designed with features encouraging social interaction, e.g., the NatureCollections app, which had a ‘My Friends’ screen (Kawas et al., 2019); Project Noah, a social media site for sharing photographs and observations (Zydney & Schaen, 2018); and ABBOT, which shared material when two devices were in close proximity (Delprino et al., 2018). Social interactions such as through emergent gameplay, collaboration, sharing content could also occur incidentally, as with the Ambient Birdhouse (Soro et al., 2018), Eko (Caiola et al., 2023), and Nature Tour application (Rikala & Kankaanranta, 2014), respectively.

Several technologies allowed for creative manipulation of nature data, but only one study made use of nature sounds (Caiola et al., 2023). Most children’s creativity was reported as nature photography, which at times was driven by an appreciation of natural beauty and desire to communicate its appeal. In one case, creative extensions were made including looped video and layered photography, using commercially available smartphone applications (Land et al., 2020). Imaginative engagement with images and environment was also facilitated by augmented reality that promoted story-telling (Kumpulainen et al., 2020). Cameras were also described in one study as a boundary object that led children to negotiate the unknown, and ask questions about what they were photographing (Chambers, 2018).

Similarly, photography often led children to notice nature that they hadn’t recognized before (Bertagnoni & Galheigo, 2021). Noticing nature occurred when children explored familiar and unfamiliar surroundings, and provided opportunities to learn about new observations and experiences, e.g. “Do you know that I had never seen this tree properly? And I come here every day” (Bertagnoni & Galheigo, 2021, p. 14). Alternatively, outdoor noticing (specifically, listening) could be prompted through learning to identify bird calls in an indoor environment (Soro et al., 2018).

Noticing was often a gateway to deeper reflection about nature, prompting children to enact labours of care through multisensory experiences of nurturing species they noticed (Renshaw et al., 2021), and caring for the surrounding environment through pro-environmental behaviours (Harwood et al., 2019). Children also experienced a heightened sense of care after taking part in science-based activities that raised the importance of care for other species (Carson et al., 2021). In one case, an educator extended children’s expression of care into political messaging that was communicated through an art exhibition of photography and radio interviews with children (Jones, 2022). The theme of care also emerged from the relational framing of many of the papers contributing to it. These studies prompted children to reflect more deeply on their relationship with other species and environments

and treated nature as another agentic being on the same level as the people involved.

7. Discussion

This scoping review of research on child-nature-technology interactions shows trends and gaps across methodologies (RQ1), approaches to incorporating child agencies into research and design (RQ2), and in terms of technologies used and subsequent nature experiences (RQ3). While we cannot make claims as to the effectiveness of different approaches or technologies in enhancing children’s nature connection, this review makes contributions to enhance future research rigour, children’s voice in research and design, and novel design pathways. These include recommendations regarding reporting and study design (7.1), for supporting children’s agency, including through speculative design work (7.2), and for the design of technologies that enhance children’s connection to nature by embedding these within relations and engaging children’s imagination and wonder (7.3).

7.1. Recommendations for reporting and study design

A key recommendation to ensure accurate reporting in child-nature-technology research, is to provide more specific detail regarding the ages of participant cohorts. This will help practitioners understand the applicability of the research to the cohorts of children they work with and implications for future research.

Questions arose as to how recruitment was approached and to what extent or in what ways diverse representation was achieved. While child-nature-technology research increased over the last ten years, most research was carried out in the U.S.A. and in urban settings, and there was little reported ethnic diversity. It is recommended that initiatives aimed at facilitating nature experiences are inclusive and meaningful for a diverse range of children, encompassing multiple contexts and intersecting aspects of identity. Exploring child-nature-technology interactions in diverse communities offers opportunity for insight into alternative environmental relationships and innovative design approaches.

A further point relates to the need for detailed accounts of participants’ activities, especially children’s. Specific accounts of what participants did will help readers understand the methodology and the degree of participation. We encountered some rich accounts of the ways in which children experienced nature, but at times these accounts were sparser or focused on researcher reflections rather than children’s experiences. We acknowledge disciplinary and theoretical variations in research reporting; nevertheless, we pose these provocations with mindfulness towards what may be useful for future researchers to consider when engaging with this area of inquiry.

Few studies evaluated the effectiveness of digital interventions in enhancing child-nature connections. Consequently, an area of consideration is how to measure improved nature connection with this cohort. One recommendation is to consider the use of validated measures of nature-connectedness. A notable challenge here is that there aren't many that would be appropriate for pre-literate/pre-verbal participants, though methods such as the biophilia interview (Rice & Torquati, 2023), and others like it (Beery et al., 2020), could be considered (see: Chawla, 2020). That said, there are opportunities to extend upon the methods already in use to produce pre-/post-tests of digital interventions. Similarly, multiple frameworks exist that provide suggestions about how to determine if an experience that a technology facilitates might be generating child nature connection, (e.g. Chawla, 2020; Giusti et al., 2018; Harvey et al., 2020; Lumber et al., 2017).

Few papers (3) attempted to define what nature means in the context of the research. Two of these works were grounded in relational theories and constructs (Harwood et al., 2019; Kumpulainen et al., 2020) viewing nature as an agentic, interconnected network of humans, other living species, the environment, and technology that was introduced in the research. The other engaged with a poetic framing of nature aligned with its focus on aesthetics (Sesigur & Edeer, 2021). The explicit early framing of nature in these papers was a useful context that applied to their research design, analysis, and the positioning of the technology. However, in the remaining 23 papers, nature was not explicitly defined as to how it was understood by the research team, and therefore didn't translate to the research design. We consider this a missed opportunity to bring a diversity of understandings of what nature means to the body of literature, which can be operationalized in technology and research design.

Furthermore, there is a need to better understand how technologies are integrated into everyday life. Future study designs might include field research that explores how families and schools mediate access to nature, and digital interventions released 'into the wild'.

7.2. Recommendations for designing with children

While this review reflected a trend towards participatory design methods, there was often inconsistency around the extent of child agency. While some studies actively engaged children to enhance their voices and included them in the co-creation of the research itself (e.g. Bertagnoni & Galheigo, 2021; White, 2015), others did not conceptualise or justify child agency in the research and design and reportedly offered children a smaller role (e.g. Bates, 2020; Gennari et al., 2019; Rikala & Kankaanranta, 2014).

Of interest is how children were supported by other adults, primarily family but also educators. This included facilitating access to nature and technology use, though overall, it is not clear from the reporting whether this facilitation supported child agencies or overrode them. Only one paper investigated these relations in-depth, including how these impacted technology use in the home (Kawas et al., 2021), though several explored technologies situated within families (e.g. Paulsen et al., 2021; Soro et al., 2018). Greater documentation of activities will aid assessment of whether child agencies are respected or encouraged.

To ensure children's active participation in research and design, researchers should strive for consistency in the conceptualization and justification of agency in participatory approaches. This could involve developing clearer guidelines for engaging children in co-creation and paying attention to the roles of adults in facilitating access to nature and technology. On this note, it is essential to carefully consider engagement with children and families. This includes reflection on children's rights, relating to participants in ethical and meaningful ways (Dobson, 2020), and considering the positioning of adults in research about children's lives, with a mindfulness towards relationality and power dynamics (Wiedenman, Lee, & Hunleth, 2023).

From a design perspective, children can drive co-creation of technologies for nature connectedness through co-speculative design

methods, giving them the opportunity to be design partners (Druin, 2001). While there is evidence of the value of speculative design and fictional inquiry with children (Andersen, 2013; Cumbo et al., 2020) there is no evidence in this review of it being applied to the design of prototype technologies with this age group (though these two studies make use of narrative in their technologies: Gennari et al., 2019; Kumpulainen et al., 2020). Speculative design methods could open opportunities to explore with children as they drive the ship through their imagination, and researchers guide via provocations. This review suggests the usefulness of appropriating existing technologies: can we repurpose existing narrative-based/storying technologies for a speculative method?

7.3. Design directions for future child-nature technologies

The technologies covered by this review were dominated by cameras due to their accessibility and ease of use by children for social and pedagogical research. Within prototype technologies there was greater diversity of interaction to include tangibles and, newly, sound. Technologies that offered both data collection and manipulation features provided outdoor and indoor activity, respectively. Data manipulation allowed children to derive new meaning from nature data, whether that was in the form of learning about species and environments, or creative exploration. Meaning making through data manipulation could further explore emotional attachments to place, one of the "pathways to nature connectedness" (Lumber et al., 2017).

Most technologies fell into the instructional, exploratory, and contributory types identified in an earlier review (Anggarendra & Brereton, 2016). An addition to this may be 'imaginary' technologies, as demonstrated by the MyAR Julle application, which encouraged imagining in nature (Kumpulainen et al., 2020). Learning, imaginative, and exploratory experiences were strongly related to children's noticing of nature and from there, to care. We note that social and playful experiences could be both designed for or emerge naturally from co-located use of technologies.

Opportunities to enhance young children's nature connection have arisen that are not present in the technologies in this review but are suggested by existing nature connection frameworks. Nature data collection for aesthetics and learning was well-represented in the review; less so, nature data manipulation for creativity and imagination. Agency, curiosity, daring and experimentation can be applied to digital play (Murcia et al., 2020), as well as nature play and exploration. Children's creativity and imaginative play may be enriching ways of fostering nature connection - in fact, these demonstrate their own ways of connecting with the natural world unconditionally. Similarly, digital technologies that build children's competencies and agencies while in nature can also foster a sense of connection (Chawla, 2020).

Given children's reliance upon adults to access nature, this also suggests a role for relational technologies that fit within families and educational contexts, particularly if they are aimed at building adults and children's positive nature orientation (Van Truong et al., 2023). Designing in the network of relations (Dema et al., 2020) also produces opportunities to embed interventions in children's everyday lives, to make it part of their lived experiences, and to expand the influence of the intervention.

Other forms of relational technologies, and ones missing from this review, include those that allow children to experience nature from other perspectives - something that is currently being explored for other age ranges and draws heavily upon speculative design (Webber et al., 2023). This also opens the possibility of exploring temporal and physical scales. For instance, children could engage with technologies that encourage slowing down or noticing through a supported phenomenological sensibility (Rodgers et al., 2023), seasonal changes, cyclic plant and animal biological processes; listening to and observing nature slowly and joyfully. Certainly, listening takes time and sound offers novel gateways to fostering noticing and intimacy with nature (Caiola et al.,

2023; Portocarrero et al., 2015; Soro et al., 2018). Some potentialities also exist within commercial technologies that can bridge different temporal rhythms (i.e., nature cameras, Vella, Ploderer, & Brereton, 2021), while others are yet to be explored - what mysteries await underground, or in the air, at the scale of the very small or tall? A challenge will be maintaining young children's attention to natural events over time, suggesting the need for research about what engages them in sustained or repeated activities.

Finally, while very young children may be considered too young to be effective in conservation efforts, their enthusiasm may make up for it. Fostering nature competencies at an early age, as well as their sense of contribution and positive action will aid them when they become more aware of the climate challenges that are facing them. A move from individual experiences towards a focus on collective experiences can promote social connection, co-creation and collaboration. The skills they will need: to collaborate, problem-solve, understand complex systems, and treat others with care - including more-than-human others (De La Bellacasa, 2017) can be designed for, with examples drawn from nature. Chawla's (2020) identification of actions encouraging constructive hope and action overlaps with that of nature connection. This framework could be applied, for example, to find ways to support early childhood participants to feel a part of collectives that protect or study nature or learn what they can do as individuals. In doing so, they can grow empowered to care for the world ethically, responsively, and effectively (Taggart, 2016), including developing their own sustainable behaviours (Barrera-Hernández et al., 2020).

Relatedly, considering how to leverage the benefits of learning in the natural world (Kuo et al., 2019) without detracting from it with technological fascination is key and may unlock new kinds of educational technologies. Directions are indicated from research on outdoor digital play (Hitron et al., 2018), though activity outdoors does not directly equate to nature connection. However, even within urban environments with challenging social contexts, children can be encouraged to appreciate natural features and find their voice, with minimal intervention (Bertagnoni & Galheigo, 2021). Designing digital technologies that promote awe and wonder and simulate natural phenomena could allow children to explore, interact, and connect with nature in more immersive ways. Importantly, technologies that generate cyclic movement between outdoor and indoor activities, of which there were several reviewed, could also consider how to augment children's imaginings of their current world, and how they might sustainably redesign its future.

8. Limitations

This review has several limitations. We excluded literature prior to 2010 and non-English language publications. The latter resulted in a collection of papers dominated by Western, Educated, Industrialized, Rich, and Democratic (WEIRD) countries. We did not include grey literature or non-peer reviewed papers. However, we did examine reference lists of the included papers, which identified two additional articles for inclusion in data extraction.

9. Conclusion

This scoping review provides an overview of research with child-

nature-technology interactions in the last ten years. It offers directions for future studies on child-nature connectedness and technology design to ensure that we can best identify the effectiveness of different approaches and extend child-centred participation, as well as explore novel approaches that align with nature connection frameworks. Specifically, recommendations are posed for more granular reporting around participants, procedures, and theoretical bases for research, and measuring impacts of digital interventions on children's nature connectedness. Novel design directions encompass 'imaginary' technologies that engage with children's personal meaning-making, relational design with adults and the more-than-human, designing for children's noticing across diverse time and space scales, and exploring how digital technologies can support the competencies required of them in our swiftly changing world. In doing so, we aim to promote a cohesive approach that moves beyond the idea of nature as a resource to be consumed solely for the sake of human health and wellbeing. Instead, we suggest a relational design approach that promotes deeper connection with nature, recognizes the interdependence between children and the natural world, and highlights their role in environmental conservation. This approach will help to strengthen child-nature-technology research and the design of technologies that move children from nature experiences to nature connection.

Selection and participation

This paper is a literature review from existing studies. Therefore, there were no participants involved directly in this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used perplexity.ai and ChatGPT to suggest more concise sentences, in the last stages of writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

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Appendix A

Keywords for Database Search

children OR child OR student* OR baby OR babies OR infant OR infants OR toddler* OR "early childhood" OR preschool OR kindergarten OR playgroup OR "elementary school" OR "nursery school" OR "early years" OR "early learning" OR "junior primary" OR "primary school" OR "pre-school"
AND.

“nature experiences” OR “nature connection” OR “nature connectedness” OR “connectedness to nature” OR “nature relat*” OR “nature stewardship” OR “nature affinity” OR biophilia OR biophobia OR “nature exploration” OR “nature play” OR “with nature” OR “for nature” OR “ecological identity” OR “environmental identity” OR “care for nature” OR “interest in nature” OR “nature-based” OR “of nature” OR “nature-centred” OR “time in nature” OR “being in nature” OR “human-nature” OR “nature interactions” OR “nature engagement” OR “more-than-human”

AND

technol* OR “app” OR application OR coding OR code OR software OR “screen time” OR screen* OR digital OR videogame* OR “video game” OR computer* OR laptop* OR “PC” OR “Mac” OR iPad* OR smartphone* OR “mobile phone” OR “smart phone” OR “AR” OR “augmented reality” OR “VR” OR “virtual reality” OR “XR” OR “extended reality” OR robot* OR “mixed reality” OR hologra* OR “social media” OR “IOT” OR “internet of things” OR device* OR prototype* OR wearable OR “smart toy” OR “ICT” OR “smart home” OR “smart display” OR sensor* OR camera* OR acoustic OR internet OR “API” OR “browser” OR webpage OR “web page” OR web OR “parental controls” OR smarthome OR “familial controls” OR “3D print*” OR photo* OR photograph*

AND NOT “secondary school students” OR “high school students” OR “university students” OR “college students” OR “higher education” OR “senior students”.

Appendix B

Data Extraction Tool

1. Coder initials:
2. Title of paper:
3. Aim of study:
4. What type of **methodology** do they use?
 - Quantitative
 - Qualitative
 - Mixed method
5. Briefly **name and describe** the **methods** being used. Include detail about role of non-researchers.
6. If the authors engage with any **theories or constructs** to frame their research, please note them here
7. Who were the **study participants**? *Tick all that apply.*
 - Children
 - Educators
 - Family members
 - Other: _____
8. **How many child participants** aged 0–8 years were there? Note how many were older than this as well if info is there.
9. What were the **child participants’ ages**? Include any statistics that the paper reports.
10. If children’s **genders** are reported, please supply that information as it is reported.
11. **Country** the research was carried out in
12. Children’s **ethnicity** or **indigenous status** if noted
13. **What kinds of nature** do the children interact with in the study? *Tick all that apply.*
 - human-managed spaces with natural features (e.g. parks, gardens)
 - pockets of wild nature in an urban setting
 - ‘wild nature’ outside of urban settings e.g. forests farms
 - Other: _____
14. Add detail or quotes here that **describe the natural space the research was carried out in**
15. If the authors attempt to **define ‘nature’** please provide more **detail or quotes** here
16. **What kind of technology** do the children use? *Tick all that apply.*
 - Acoustic recorders
 - Augmented reality (AR)
 - Cameras, handheld
 - Cameras, worn
 - iPad or similar
 - Prototype technology
 - Smartphone application
 - Social media
 - Computers (e.g. in a classroom)
 - Other: _____
17. Can you **please describe the technology** in more detail?
18. What is the **aim of the technology** in the context of this study? *Tick all that apply.*
 - to encourage children to explore nature to help children learn about nature
 - to help children notice nature
 - to help children understand sustainability in a natural context
 - to prompt reflection on nature experiences or connection to nature
 - to provide information to the researchers about children’s nature experiences
 - to provide information to the researchers on a topic unrelated to nature
 - the aim is not stated/it’s not clear
 - Other: _____

19. What types of nature experiences did the children have as a result of using the technology? *Tick all that apply.*
- appreciation of natural beauty/aesthetics
 - expanding their knowledge of nature
 - exploration and/or discovery
 - building nature-related skills
 - imaginative experiences in nature
 - physical activity
 - playful or gameful experiences in nature
 - positive bonds with plant/animal species
 - social nature experiences with other children or adults
20. Please provide more **details or quotes** to illustrate this
21. How much **agency** did the children have in terms of how they **used the technology**? *Tick all that apply.*
- They were explicitly told what to do with the technology by the researchers
 - They were explicitly told what to do with the technology by other adults
 - They were given rough guidelines of what to do with the technology by the researchers
 - They were given rough guidelines of what to do with the technology by other adults (e.g. educators, family)
 - They were not told what to do with the technology, only shown how to use it, by the researchers
 - They were not told what to do with the technology, only shown how to use it, by other adults
 - That information was not supplied
 - Other: _____
22. If possible, provide **more detail or quotes** here that illustrate **how the children's technology use was framed or prompted**
23. What **role** do children play in the research? *Tick all that apply*
- Children supply data about their experiences in nature
 - Children supply data about their nature connectedness
 - Co-design user - children use a technology while adults record data on this use
 - Co-design tester - children test a technology and provide their experiences
 - Co-design informant - children are involved in the early stages of a design process
 - Co-design partner - children contribute to every stage of a design process as equal partners
 - Children lead and/or provide direction about activities
 - Other: _____
24. If possible, supply **detail or quotes** that illustrate this
25. What role do **adult participants** play in the research? *Tick all that apply*
- Providing data to the researchers about children's experiences
 - Guiding or leading or facilitating technology use
 - Taking children to natural spaces
 - Educating children/linking experiences to pedagogy
 - Other: _____
26. Please provide **more detail about the role of adult participants** including if they are **educators, family members, etc.**
27. If there is **anything else** you think worth capturing, please use this space

Appendix C

Table 1

Complete list of included papers with research setting, methods, and participant details.

Papers that reported on evaluations of technologies designed to encourage nature connection or experiences are indicated with a *

Reference	Location	Setting	Methods	Age of child participants	Role of non-researcher adult
Bates (2020)	U.K.	School grounds	● Qualitative ● Interviews ● Video tour and photographic workshops ● Mostly child-led photography and videography using wearable (digital) and hand-held (non-digital) cameras, iPads	5–6 years and 7–8 years	● Providing data to researcher
Bertagnoni and Galheigo (2021)	Brazil	Parks and various outdoors urban sites	● Qualitative ● Workshops ● Child-led photographic tour and photovoice (interview) using handheld digital cameras	7–8 years	● Taking children outdoors
Bruni et al. (2017)*	U.S.A.	National forest; indoors; and various unknown sites including outdoors	● Observation ● Quantitative ● Questionnaires (Implicit Association test - Nature) ● Quantitative analysis	5–6 and 7–8 years	● Taking children outdoors ● Guiding or leading or facilitating technology use

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Table 1 (continued)

Reference	Location	Setting	Methods	Age of child participants	Role of non-researcher adult
Caiola et al. (2023)*	Italy	Public park	• Mixed methods • Interviews • Focus group • Questionnaires • Researcher reflection	5–6 years and 7–8 years	• Guiding or leading or facilitating technology use
Carson et al. (2021)	New Zealand	Seashores	• Mixed methods • Surveys (qualitative and quantitative measures) • Interviews	7–8 years	• Taking children outdoors • Guiding or leading or facilitating technology use • Taking children outdoors
Chambers (2018)	U.S.A.	School grounds	• Qualitative • Child-led photography using handheld digital cameras • Collection of artefacts (journals, photography, and text that accompanies photography) • Observation Field notes	7–8 years: grade levels given “groups of students in grades 3–5” (children are typically aged between 8 and 11 years in these grade levels)	• Taking children outdoors
Delprino et al. (2018)*	Italy	School garden	• Qualitative • Interviews • Focus groups • While there was also child-led photography using a prototype device (ABBOT) – these photos were not used in the analysis	3–4, 5–6, and 7–8 years	• Taking children outdoors • Providing data to researcher
Gennari et al. (2019)*	Italy	School garden	• Mixed methods • Interviews • Focus groups • Questionnaires • Observation	3–4, 5–6, and 7–8 years	• Providing data to researcher
Green (2017)	U.S.A.	A nature camp located in sub-alpine forested hills	• Qualitative • Child-led tour of natural environments (videography using wearable cameras) • Interview using videos to aid recall • Observation	5–6 and 7–8 years	• Taking children outdoors
Green (2022)	U.S.A.	Fields and a boreal forest at a bird migratory refuge; botanic gardens; school and public playgrounds; suburban streets; ‘hills’ outside the city	• Qualitative • Mostly child-led tours of natural surroundings (videography with wearable cameras) • Observation using notes, photographs, and audio recordings of conversations	3–4 years	• Taking children outdoors
Harwood et al. (2019)	Canada	Woodland located on a university campus	• Qualitative • Tours of natural environments (videography by children using wearable cameras) • Observations including field notes, photographs	3–4 years	• Taking children outdoors • Educating children
Jones (2022)	U.K.	Sancreed Beacon; moors and woodland	• Qualitative • Child-led photography using handheld cameras • Text produced by children to accompany their photographs • Interviews	7–8 years	• Taking children outdoors • Educating children
Kawas et al. (2019)*	U.S.A.	An outdoor garden and a local park	• Mixed methods • Interviews • Observations including field notes • Child-led photography using smartphone application • Collection of artefacts	7–8 years	• Taking children outdoors • Guiding or leading or facilitating technology use
Kawas et al. (2021)*	U.S.A.	Various outdoor settings including walks around neighbourhood, beaches, hikes, etc.	• Qualitative • Interviews with parents and children	5–6 years and 7–8 years	• Taking children outdoors • Providing data to researcher
Kumpulainen et al. (2020)*	Finland	The neighbourhood surrounding a school; schoolyards; woodlands; located in eastern part of Helsinki	• Qualitative • Interviews • Observation using field notes • Analysis of digital stories created by children using prototype technology	7–8 years	• Guiding or leading or facilitating technology use
Land et al. (2020)	Canada and Australia	Haro Woods in British Columbia, Canada, and Cruickshank park in Melbourne, Australia.	• Qualitative • Child- and researcher-led photography, videography, and sound collection using smartphones	3–4 years: grade levels given (“preschool and kindergarten-aged children” – children are typically 3–4 years of age)	• Taking children outdoors

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Table 1 (continued)

Reference	Location	Setting	Methods	Age of child participants	Role of non-researcher adult
			• Informal conversations between children, educators, researchers		• Guiding or leading or facilitating technology use • Educating children • To gain consent for child participation
McClain and Zimmerman (2016)*	U.S.A.	Shaver's Creek Environmental Center located near Rothrock State Forest, U. S.A.	• Qualitative • Observation using video recordings of children interacting with the device (iBook)	7–8 years	
Paulsen et al. (2021)*	U.S.A.	Gardens, parks and trails in neighbourhoods	• Mixed methods • Interviews • Surveys	5–6 and 7–8 years	• Taking children outdoors • Guiding or leading or facilitating technology use • Taking children outdoors
Renshaw et al. (2021)	Australia	Home gardens	• Qualitative • Interviews • Observation • Children's videography with "digital devices"	5–6 years	• Taking children outdoors
Rikala and Kankaanranta (2014)*	Finland	Field trips to places with natural features (no further detail).	• Qualitative • Interviews with educators • Educator observations	5–6 years	• Taking children outdoors • Providing data to researcher • Guiding or leading or facilitating technology use • Educating children • Taking children outdoors • Educating children
Sesigur and Edeer (2021)	Turkey	Arboretum	• Qualitative • Interviews • Observation • Diary reflection (both researcher and participant)	7–8 years	• Educating children
Soro et al. (2018) *	Australia	Home environment; also report on experiences while at school	• Qualitative • Interviews • Diary (adult entries)	3–4, 5–6, 7–8 years	• Guiding or leading or facilitating technology use
White (2015)	New Zealand	Mount Mauao and Summerhill Farm	• Qualitative • Child-led photography using handheld digital cameras (photovoice) • Interviews • Diary reflection (educators)	3–4 years	• Taking children outdoors • Educating children
Zydney and Schaen (2018)	U.S.A.	Schoolyard	• Qualitative • Child-led photography using iPads • Observations	6–7 years: grade levels given ("first grade")	• Guiding or leading or facilitating technology use • Educating children

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